

Presentation Layer (User Interface Layer)

Web and Mobile Application.
Login and registration screens.
Interactive map displaying real-time boat locations.

Application / Business Logic Layer

Process GPS data received from tracking chips.
Detect when a boat leaves a defined area or loses signal.

Data Access Layer

Send and retrieve data from the database.
Ensure data integrity and consistency.
Manage SQL queries and stored procedures.
Provide a secure interface for reading/writing data.

Database Layer

Boat information and unique IDs.
GPS coordinates and movement history.
User accounts and permissions.
Alerts, reports, and system logs.

The Layered Architecture was chosen because it makes the system organized, easy to manage, and clear to understand. Each layer has its own job — the user interface, system logic, data access, and database — which helps in maintenance, security, and future updates.

app used: canva